

IGNITE Your PASSION FOR DIY With This DVD

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Arduino 1.6.8

Arduino has been around for quite some time in the Internet of Things zone. Arduino boards enable computers to assess and interact with the surroundings by processing data from sensors. Being an open source platform, the programs are modifiable by anyone who wishes to work on it.

Equipped with open source Arduino software (IDE), writing codes and uploading to the board is easy. The environment is written in Java and based on Processing and other open source software. It can run on all popular platforms such as Windows, Mac OS X and Linux.

Kubuntu

Kubuntu is another popular flavour of operating systems based on the Ubuntu platform. It uses KDE Plasma Desktop instead of the Unity graphical environment. All the underlying systems are the same and Kubuntu shares the same package repository as Ubuntu.

Kubuntu has an illuminated Plasma Desktop that is fully customisable. It has been designed in such a way that you never feel the difference in interface, especially if you are familiar with Windows operating system. There are some useful and beautiful widgets in this package, too. Desktop effects are integrated in the standard installation and enabled by default.

The Ubuntu and Kubuntu use the same sources, which lets any application to be used, installed and run on either of these.

Digital Circuit Design

Digital Logic Design is a software tool for designing and simulating digital circuits. It provides digital parts ranging from simple gates to arithmetic logic

Some popular resources

FileZilla. FileZilla is a cross-platform graphical FTP, SFTP and FTPS file management tool for Windows, Linux, Mac OS X and more. It is a bundle of many intuitive tools. FileZilla helps you quickly move files between your computer and Web server. There is a provision to use the speed monitoring tools if you want to go into advanced mode. While FTP may seem outdated, it's actually a very reliable way to transfer large files or groups. FileZilla can deploy multiple simultaneous connections to speed up file transfers.

Sweet Home 3D. Sweet Home 3D is an interior design application that helps you to quickly draw the floor plan of houses, find a better arrangement for furniture in it, and visit the results in 3D. Features are:

- Draw walls and rooms upon the image of an existing plan, on one or more levels
- Drag and drop doors, windows and furniture from a catalogue onto the plan
- Update colours, texture, size and orientation of furniture, walls, floors and ceilings
- View all changes simultaneously in the 3D view from any view point
- Create photorealistic images and videos with various light sources
- Import additional 3D models and export the plan at various standard formats

units and state machine. Digital Circuit Design helps its users to convert circuits into easily reusable models.

A module may be used to build more complex circuits like a central processing unit. The software is equipped with functionalities that can show the working circuit using output parts like light emitting diodes, 7-segment display as well as cathode ray tube and digital oscilloscope provided in the software.

Wiring

Wiring is both a programming environment and an electronics prototyping input/output board for exploring tangible media. This is again something similar to Arduino and can be used to teach anyone the wonderful world of computer programming and prototyping.

The software illustrates the con-

cept of programming with electronics while exploring the physical reach of hardware control. This process is called sketching like in Arduino. The user can create creative prototypes, explore lots of ideas very quickly, select interesting ones, and make this a repeating process.

Papilio DesignLab

This DesignLab works with Windows and Linux to enhance the experience of drawing and debugging your own circuits. Like all IDEs, Papilio DesignLab lets the user develop the Hardware Description Language just by typing code into the IDE, which in turn generates the file that is loaded into the FPGA. The Papilio IDE includes a text editor, libraries, revision control, syntax highlighting and a compiler. Its unique selling point is the drag-and-drop interface for designing FPGAs using the Arduino IDE. The tool is limited for use with Papilio boards only.

Qucs

Qucs is an integrated circuit simulator that helps you set up a circuit with a graphical user interface and simulate the large-signal, small-signal and noise behaviours of the circuit.

Once the simulation is done, you can view the simulation results on a presentation page or window. Qucs is currently developed by open source enthusiasts under the GNU/Linux OS, using the standard auto tools with no special effort to support other operating systems. However, Qucs can be successfully compiled and run on Windows, Solaris, NetBSD, FreeBSD, MacOS and Cygwin. **EFY**

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